

Q-13: Consider a utility maximization problem. If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^N$, then examine the properties satisfied by the Walrasian demand function $x(p, w)$. Examine the situation when the sufficient condition of UMP may be violated.

Ans:

Consumer maximizes his utility level by selecting a bundle x , subject to his budget constraint:

$$\begin{aligned} \max_{x \geq 0} u(x) \\ \text{s.t. } p \cdot x \leq w \end{aligned}$$

If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^N$, then the Walrasian demand $x(p, w)$ satisfies: (properties of Walrasian demand)

① Homogeneity of degree zero:

$$x(p, w) = x(\alpha p, \alpha w) \text{ for all } p, w \text{ and for all } \alpha > 0.$$

That is, the budget set is unchanged.

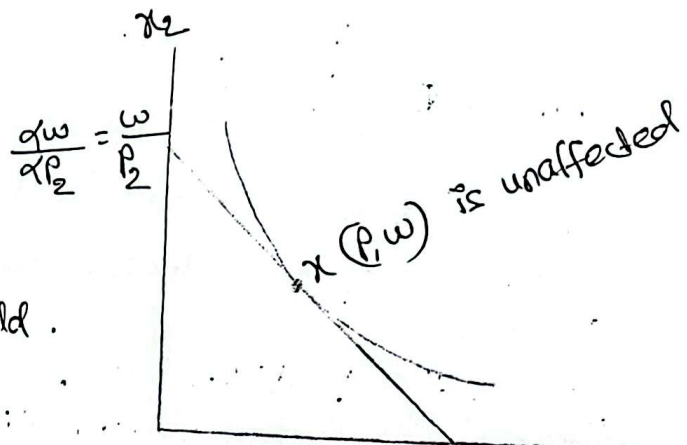
$$\{x \in \mathbb{R}_+^N : p \cdot x \leq w\} = \{x \in \mathbb{R}_+^N : \alpha p \cdot x \leq \alpha w\}.$$

We don't need any assumption on the preference relation

LNS: local non-satiation

to show this. We only rely on the budget set being unaffected.

Note that, the preference relation can be linear and homog (0) would still hold.



[In the fig. budget line $B_{p,w}$, with intercepts w/p_1 and w/p_2 and illustrates how a common increase in price and wealth produces budget line $B_{p',w'}$ with exactly same intercepts w'/p'_1 and w'/p'_2 as budget line $B_{p,w}$. $B_{p,w}$ and $B_{p',w'}$ exactly coincide.]

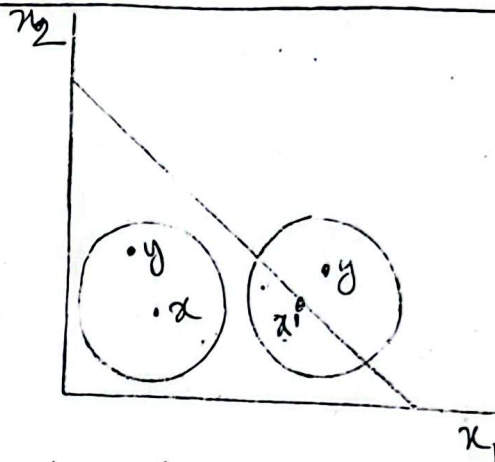
② Walras' Law:

$$p \cdot x = w \text{ for all } x = x(p, w)$$

It follows from LNS: if the consumer selects a Walrasian demand $x \in x(p, w)$, where $p \cdot x < w$, then it means we can still find other bundle y , which is ϵ -close to x , where consumer can improve his utility level.

If the bundle the consumer chooses lies on the budget line, i.e. $p \cdot x' = w$, we could then identify bundles that are strictly preferred to x' , but these bundles would be unaffordable to the consumer.

For $x \in x(p, w)$, there is a bundle y , ϵ -close to x , such that $y \succ x$. Then $x \notin x(p, w)$



उपगत Walras's Law - निम्नलिखित सत्यता का प्रमाण है, $p \cdot x = w$

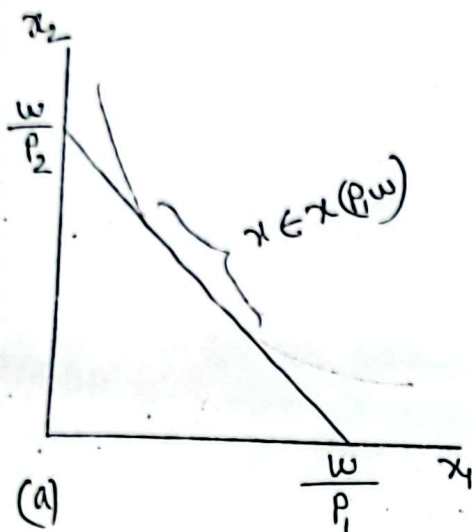
Walras's law follows from LNS. Indeed, if the consumer were to select a bundle that lies strictly inside the budget set (so that he is not exhausting his wealth, point x), we could find another point y at ϵ -distance from bundle x that is strictly preferred by the individual to bundle x . In this case, the initial bundle x would not be utility maximizing because there are other bundles that are still affordable on the budget line x' and we could still identify bundles that are strictly preferred to x' .

such as those to the However, these bundles would be unaffordable to the consumer and thus can't be utility maximizing. In summary, utility maximizing bundles, such as x' , must lie on the budget line, thus exhausting all wealth.

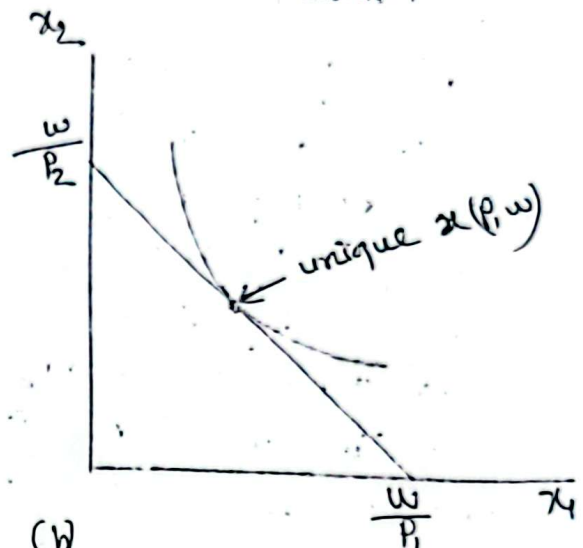
② Convexity / Uniqueness:

① If the preferences are convex, then the Walrasian demand correspondence $x(p, w)$ defines a convex set, i.e. a continuum of bundles are utility maximizing. The budget line and indifference curve can become tangent along several bundle forming a convex set.

⑥ If the preferences are strictly convex, then the Walrasian demand correspondence $x(p, w)$ contains a single element. (unique bundle) unique tangency point between budget and indifference curve



(a) fig: convex preference



(b) fig: strictly convex preference

UMP: Violation of Sufficient Condition

① ~~The consumer~~ ^{not} $u(\cdot)$ is $\nearrow$ monotone:

The consumer chooses bundle A (at a corner) since it yields the highest utility level given his budget constraint.

At point A, however, the tangency condition,

$MRS_{1,2} = \frac{p_1}{p_2}$ does not hold since

the consumer would like to further decrease his consumption of x_1 .
 Monotone implies that if we increase both goods, we reach a higher utility level and that if we increase the amount of some goods, the consumer is not made worse off.

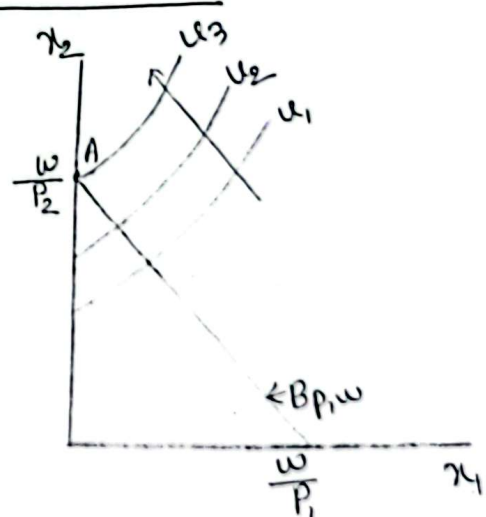
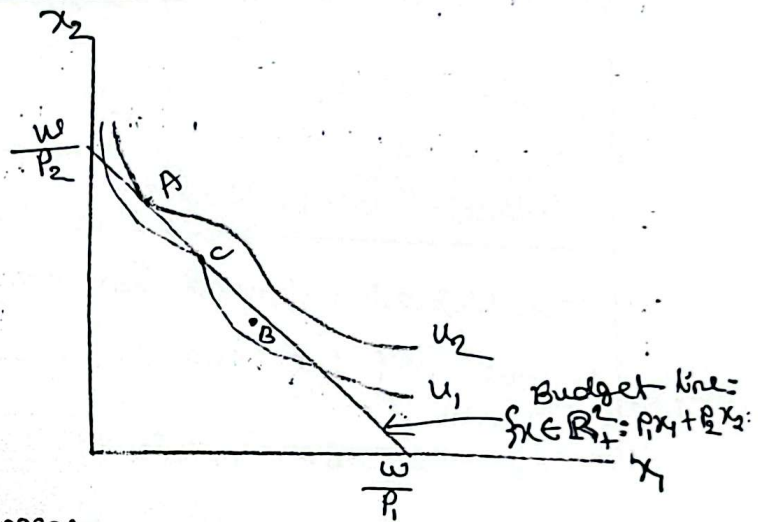


fig: corner solution

(2) $U(1)$ is not quasiconcave:

The upper contour sets (UCS) of c is not convex.

$MRS_{1,2} = \frac{p_1}{p_2}$ is not a sufficient condition for a max. A point of tangency (c) gives a lower utility level than a point of non-tangency



(B). The true maximum is at point A. Because the three requirement for necessary conditions to become sufficient condition are relatively mild (utility function being quasi-concave, monotonic and $\nabla U \neq 0$).

Q-19: Define indirect utility function. Examine its properties.

Ans:

Indirect Utility Function

The Walrasian demand function, $x(p, w)$, is the solution to the UMP (i.e. argmax).

When we insert the optimal consumption bundle into the individual's utility function, we obtain the highest utility level that the individual can achieve by solving his UMP, $u(x(p, w))$, or compactly referred to as the indirect utility function and denoted by $v(p, w)$.

Or,

~~what would~~

Indirect utility function (i.e. the highest utility level), $v(p, w) \in \mathbb{R}$, associated with the UMP, would be the utility function evaluated at the solution of the UMP, i.e. $x(p, w)$. It is the "value function" of the optimization problem.

Properties of Indirect Utility Function

If the utility function is continuous and preferences satisfy LNS over the consumption set $X = \mathbb{R}_+^L$, then the indirect utility function $v(p, w)$ satisfies:

① Homogeneous of degree zero:

Increasing p and w by a common factor $\alpha > 0$ does not modify the consumer's optimal consumption bundle, $x(p, w)$, nor his maximal utility level, measured by $v(p, w)$.

Or,

This property states that increasing market prices and wealth by the same proportion does not modify the consumer's budget set (see fig).

As a consequence, such an increase does not modify the consumer's optimal consumption bundle $x(p, w)$, nor does it affect the maximal utility level that the individual reaches in his UMP, as measured by $v(p, w)$, meaning $v(\alpha p, \alpha w) = v(p, w)$.

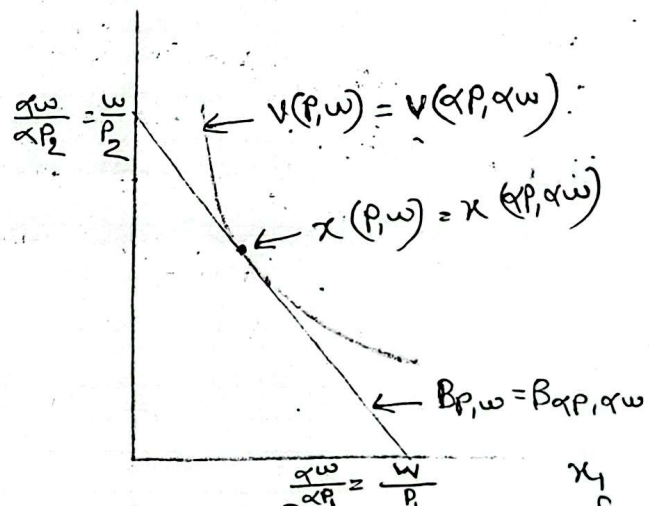


Fig: $v(p, w)$ homogeneous of degree zero.

② Strictly increasing in w : $v(p, w') > v(p, w)$ for $w' > w$.

This property states that if we increase the wealth of the individual, we enlarge the set of feasible bundles he can afford and, as a consequence, we allow him to reach an indifference curve associated with a higher utility level.

Therefore, the maximal utility level that he can obtain is strictly increasing in his wealth, as depicted in the fig. where $v(p, w') > v(p, w)$ because $w' > w$.

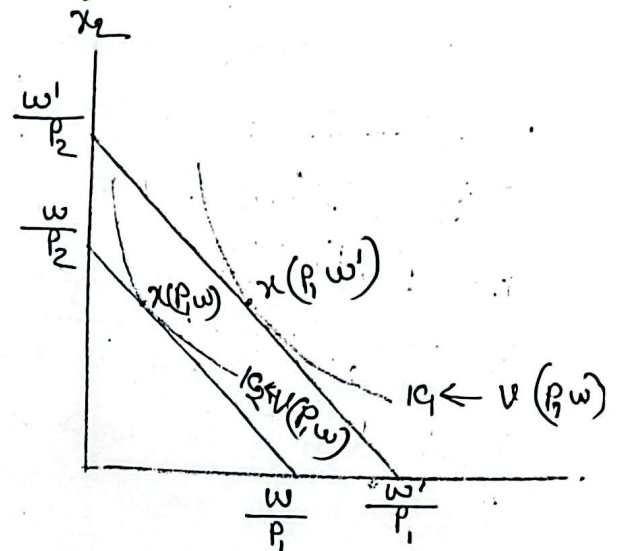
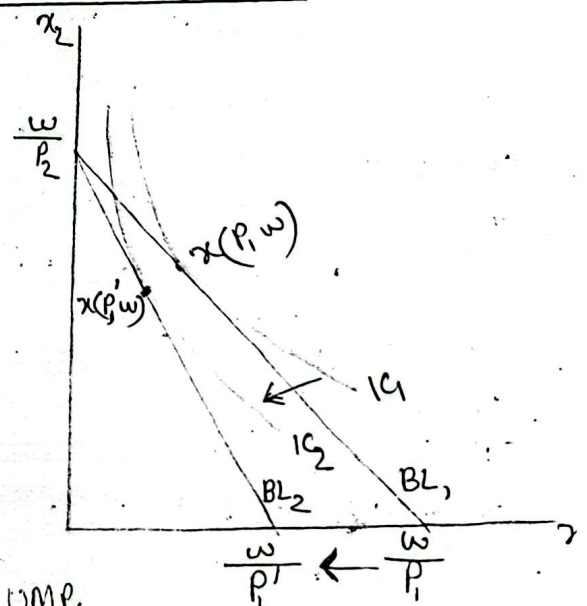


Fig: $v(p, w)$ strictly increase in w .

③ Non-decreasing (i.e. weakly decreasing) in p_k :

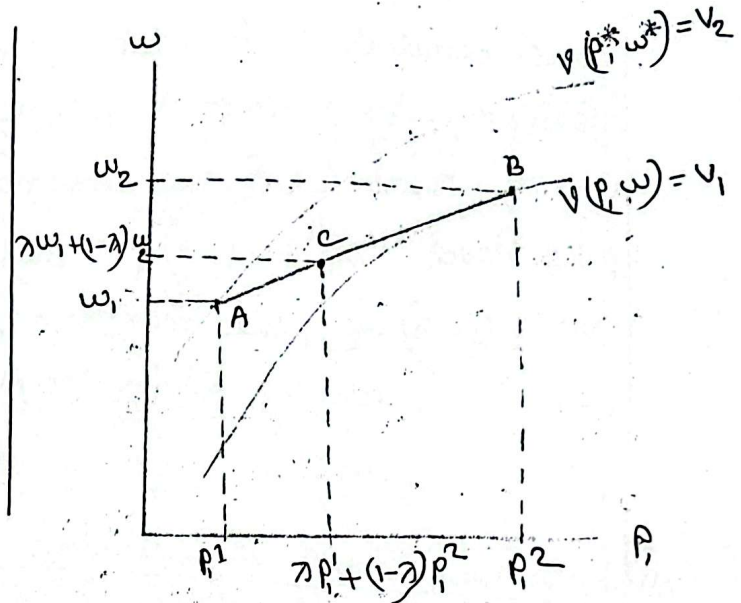
With an increase in the price of any good, the set of affordable bundles shrinks, so the individual can only reach indifference curves associated with weakly lower utility levels. Hence, an increase in the price of any good k produces a weak reduction in the maximal utility level that the individual can obtain by solving this UMP.



④ Quasi-convexity: The set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for any $\bar{v}$.

Interpretation - I:

If $(p^1, w^1) \succeq^* (p^2, w^2)$,
 then $(p^1, w^1) \succeq^* (\lambda p^1 + (1-\lambda)p^2, \lambda w^1 + (1-\lambda)w^2)$;
 i.e. if $A \succeq^* B$, then $A \succeq^* C$.



or, The set $\{(p, w) : v(p, w) \leq \bar{v}\}$ is convex for any $\bar{v} \in \mathbb{R}_+$. The figure depicts the indirect utility function with the price of good 1 on the horizontal axis and wealth level on the vertical axis. When price increases from p^1 to p^2 , this individual's also wealth must also increase in order to keep his utility level unaffected. Graphically utility level $v(p^*, w^*) > v(p, w)$ where in contrast, lower prices and higher wealth levels are associated with higher maximal utilities.

Graphically, utility level $v(p^*, w^*) > v(p, w)$

where $p^* < p$ and $w^* > w$.

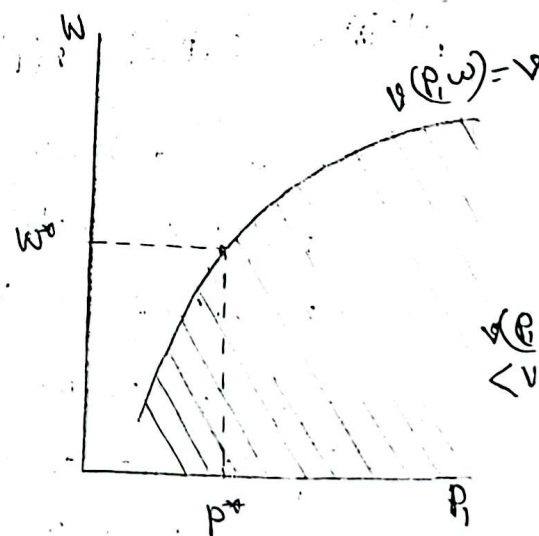
Let us now examine the quasi-convexity of $v(p, w)$. This property implies that, if the maximum utility associated with a given pair of prices and wealth (point A) is

weakly higher than the maximum utility associated with another pair of prices and wealth (point B), then the convex combination of points A and B (point C) yields a maximum utility that is weakly lower than that associated with A. That is, if the consumer faces an intermediate price level $\gamma p_1^1 + (1-\gamma) p_1^2$ and an intermediate wealth level $\gamma w_1 + (1-\gamma) w_2$, the maximum utility level that he will be able to reach by solving his UMP is lower than that at point $A \equiv (p_1^1, w_1)$.

Alternative Interpretation of Quasi-convexity

Interpretation II:

$V(p, w)$ is quasiconvex if the set of (p, w) pairs for which $V(p, w) < V(p^*, w^*)$ is convex.

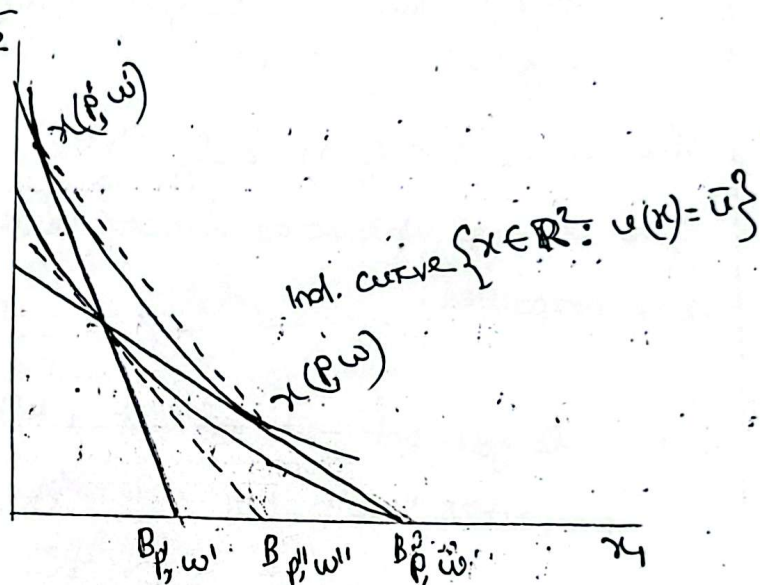


Another interpretation of Quasi-convexity

Interpretation III:

Using x_1 and x_2 in the axis, perform following steps:

- ① When $B_{p, w}$ budget then $x(p, w)$ bundle
- ② When $B_{p', w'}$ then $x(p', w')$



- ③ Both $x(p, w)$ and $x(p', w')$ induce an indirect utility of $v(p, w) = v(p', w') = \bar{u}$

- ④ Construct a linear combination of prices and wealth:

$$\left. \begin{aligned} p'' &= \alpha p + (1-\alpha)p' \\ w'' &= \alpha w + (1-\alpha)w' \end{aligned} \right\} B_{p'', w''}$$

- ⑤ Any solution to the VMP given $B_{p'', w''}$ must lie on a lower indifference curve (i.e. lower utility).

$$v(p'', w'') \leq \bar{u}$$

Q-15: Consider the Cobb-Douglas utility function:

$$U(x_1, x_2) = k x_1^\alpha x_2^{1-\alpha}$$

for some $\alpha \in (0, 1)$ and $k > 0$

i) Derive the Walrasian demand functions and examine their properties.

ii) How do you get the indirect utility function. Explain their properties. Verify Roy's identity.

iii) Solve the expenditure ^{minimization} ~~maximization~~ problem and explain the properties of the compensated demand function.

Answer:

i) Deriving Walrasian Demand Function

$$U(x_1, x_2) = k x_1^\alpha x_2^{1-\alpha}$$

$$\text{Budget constraint: } p_1 x_1 + p_2 x_2 = W$$

Setting up the Lagrangian,

$$L = k x_1^\alpha x_2^{1-\alpha} + \lambda [W - p_1 x_1 - p_2 x_2]$$

FONC requires

$$\frac{\partial L}{\partial x_1} = k \alpha x_1^{\alpha-1} x_2^{1-\alpha} - \lambda p_1 = 0 \quad \dots \textcircled{i}$$

$$\frac{\partial L}{\partial x_2} = k x_1^\alpha (1-\alpha) x_2^{-\alpha} - \lambda p_2 = 0 \quad \dots \textcircled{ii}$$

W = wealth
P = price
Walrasian demand
max. $U(x)$
s.t. $Px \leq W$

$$\frac{SL}{SN} = W - P_{x_1} \cdot x_1 - P_{x_2} \cdot x_2 = 0 \dots \textcircled{iii}$$

Dividing Eq. ① by Eq. ③,

$$\frac{K \alpha x_1^{\alpha-1} x_2^{1-\alpha}}{K x_1^{\alpha} (1-\alpha) x_2^{-\alpha}} = \frac{x_1 P_{x_1}}{x_1 P_{x_2}}$$

$$\Rightarrow \frac{\alpha}{1-\alpha} x_1^{\alpha-1-\alpha} x_2^{1-\alpha+\alpha} = \frac{P_{x_1}}{P_{x_2}}$$

$$\Rightarrow \frac{\alpha}{1-\alpha} x_1^{-1} x_2 = \frac{P_{x_1}}{P_{x_2}}$$

$$\Rightarrow \frac{\alpha}{1-\alpha} \cdot \frac{x_2}{x_1} = \frac{P_{x_1}}{P_{x_2}}$$

$$\Rightarrow P_{x_2} \cdot x_2 = P_{x_1} \cdot x_1 \cdot \frac{1-\alpha}{\alpha} \dots \textcircled{iv}$$

Substituting Eq. ④ into Eq. ③,

$$W - P_{x_1} \cdot x_1 - P_{x_2} \cdot x_2 = 0$$

$$\Rightarrow W - P_{x_1} \cdot x_1 - P_{x_1} \cdot x_1 \cdot \frac{1-\alpha}{\alpha} = 0$$

$$\Rightarrow W - P_{x_1} \cdot x_1 \left(1 + \frac{1-\alpha}{\alpha} \right) = 0$$

$$\Rightarrow W - P_{x_1} \cdot x_1 \left(\frac{\alpha + 1 - \alpha}{\alpha} \right) = 0$$

$$\Rightarrow W - P_{x_1} \cdot x_1 \cdot \frac{1}{\alpha} = 0$$

$$\Rightarrow W = P_{x_1} \cdot x_1 \cdot \frac{1}{\alpha}$$

$$\Rightarrow P_{x_1} \cdot x_1 = \alpha W$$

$$\Rightarrow x_1 = \frac{\alpha W}{P_{x_1}}$$

$$\text{Similarly, } x_2 = \frac{(1-\alpha)W}{P_{x_2}}$$

So, Walrasian demand for goods $x_1 = \frac{\alpha W}{P_{x_1}}$ and $x_2 = \frac{(1-\alpha)W}{P_{x_2}}$

The demand for every good k is increasing in wealth w , decreases in its own price, P_k , but unaffected by the price of other goods, P_l , where $l \neq k$.

Properties of Walrasian demand

① Homogeneity of degree zero $x(p, w) = x(\alpha p, \alpha w)$ for all p ; $w > 0$ and for all $\alpha > 0$.

② Walras's law, $p \cdot x(p, w) = w$

③ Convexity/uniqueness: If preferences are convex, then the Walrasian demand $x(p, w)$ defines a convex set. If preferences are strictly convex, then the Walrasian demand $x(p, w)$ contains a single element.

*+ Details $\Rightarrow$ Q-13 to 14

$$\left. \begin{aligned} x_1(p, w) &= x_1 \\ x_2(p, w) &= x_2 \end{aligned} \right\} \begin{array}{l} \text{(ଅ-ଭୋଗ)} \\ \text{ଏକସାର} \\ \text{ନିର୍ଦ୍ଦେଶ} \end{array}$$

Q.15

(ii) Indirect utility function $= v(p, w)$

Walrasian demands for good 1 and 2 are,

$$x_1(p, w) = \frac{\alpha w}{p_{x_1}}$$

$$x_2(p, w) = \frac{(1-\alpha)w}{p_{x_2}}$$

Indirect utility function:

$$\begin{aligned} v(p, w) &= k(x_1)^\alpha \cdot x_2^{1-\alpha} \\ &= k\left(\frac{\alpha w}{p_{x_1}}\right)^\alpha \cdot \left(\frac{(1-\alpha)w}{p_{x_2}}\right)^{1-\alpha} \\ &= k\left(\frac{\alpha}{p_{x_1}}\right)^\alpha \cdot w^\alpha \cdot \left(\frac{1-\alpha}{p_{x_2}}\right)^{1-\alpha} \cdot w^{1-\alpha} \\ &= k\left(\frac{\alpha}{p_{x_1}}\right)^\alpha \cdot \left(\frac{1-\alpha}{p_{x_2}}\right)^{1-\alpha} \cdot w^{\alpha+1-\alpha} \\ &= k\left(\frac{\alpha}{p_{x_1}}\right)^\alpha \cdot \left(\frac{1-\alpha}{p_{x_2}}\right)^{1-\alpha} \cdot w \end{aligned}$$

^{indirect} This utility function is decreasing in the price of both goods, increasing in the consumer's wealth, w and homogeneous of degree zero, since increasing all prices and wealth by a common factor A yields,

$$\begin{aligned} v(AP, AW) &= k\left(\frac{\alpha}{AP_{x_1}}\right)^\alpha \cdot \left(\frac{1-\alpha}{AP_{x_2}}\right)^{1-\alpha} \cdot (AW) \\ &= k\left(\frac{\alpha}{p_{x_1}}\right)^\alpha \cdot A^{-\alpha} \cdot \left(\frac{1-\alpha}{p_{x_2}}\right)^{1-\alpha} \cdot A^{-1+\alpha} \cdot A^1 \cdot w \\ &= A^{-\alpha-1+\alpha+1} \cdot k\left(\frac{\alpha}{p_{x_1}}\right)^\alpha \cdot \left(\frac{1-\alpha}{p_{x_2}}\right)^{1-\alpha} \cdot w \\ &= A^0 \cdot v = v \quad [A^0 = 1] \end{aligned}$$

Homogeneity of degree 0

Properties of indirect utility function $\Rightarrow$ see -(Q-14)

Roy's Identity:

Relationship between the Walrasian demand and the indirect utility function:

Consider a continuous utility function representing a strictly convex preference relation that satisfies LNS. Additionally suppose that the indirect utility function associated with the VMP problem is differentiable at any pair of prices and wealth. We then obtain

Roy's Identity:

$$-\frac{\frac{SV(p, w)}{SP_k}}{\frac{SV(p, w)}{SW}} = x_k(p, w) \text{ for every good } k.$$

In our example,

$$\begin{aligned} V &= K \left(\frac{\alpha}{P_{x1}} \right)^{\alpha} \left(\frac{1-\alpha}{P_{x2}} \right)^{1-\alpha} W \\ \frac{SV(p, w)}{SP_{x1}} &= \frac{\left(\alpha \cdot P_{x1}^{-1} \right)^{\alpha} \cdot K \left(\frac{1-\alpha}{P_{x2}} \right)^{1-\alpha} W}{\alpha \cdot \left(P_{x1}^{-1} \right)^{\alpha-1} \cdot \frac{d}{dP_{x1}} \left(P_{x1}^{-1} \right) \cdot K \left(\frac{1-\alpha}{P_{x2}} \right)^{1-\alpha} W} \\ &= \frac{\alpha \cdot \left(P_{x1}^{-1} \right)^{\alpha-1} \cdot \frac{d}{dP_{x1}} \left(P_{x1}^{-1} \right) \cdot K \left(\frac{1-\alpha}{P_{x2}} \right)^{1-\alpha} W}{\alpha \cdot \left(P_{x1}^{-1} \right)^{\alpha-1} \cdot \ln P_{x1} \cdot K \left(\frac{1-\alpha}{P_{x2}} \right)^{1-\alpha} W} \end{aligned}$$

In our example,

$$V = K \left(\frac{\alpha}{P_{x_1}} \right)^\alpha \left(\frac{1-\alpha}{P_{x_2}} \right)^{1-\alpha} W \quad (K = \text{constant})$$

$$= K \cdot \alpha^\alpha \cdot P_{x_1}^{-\alpha} \cdot (1-\alpha)^{1-\alpha} \cdot P_{x_2}^{1-\alpha} \cdot W$$

$$\eta_1 = - \frac{\frac{\partial V}{\partial P_{x_1}}}{\frac{\partial V}{\partial W}}$$

$$= - \frac{(-\alpha) P_{x_1}^{-\alpha-1} \cdot K \cdot \alpha^\alpha \cdot (1-\alpha)^{1-\alpha} \cdot P_{x_2}^{1-\alpha} \cdot W}{P_{x_1}^{-\alpha} \cdot K \cdot \alpha^\alpha \cdot (1-\alpha)^{1-\alpha} \cdot P_{x_2}^{1-\alpha} \cdot (\alpha-1)}$$

$$= -(-\alpha) P_{x_1}^{-\alpha-1+\alpha} W$$

$$= \frac{\alpha W}{P_{x_1}}$$

Same as,

$$\eta_2 = - \frac{\frac{\partial V}{\partial P_{x_2}}}{\frac{\partial V}{\partial W}} = \frac{(1-\alpha) W}{P_{x_2}}$$

— 0 —

(ii) Given that,
 $U^*(x_1, x_2) = kx_1^\alpha x_2^{1-\alpha}$

Solve Expenditure Minimization Problem:

Min. $p_1x_1 + p_2x_2$
 s.t. $U(x_1, x_2) \geq u$

Consider a consumer with Cobb-Douglas utility function,
 $U(x_1, x_2) = kx_1^\alpha x_2^{1-\alpha}$. We can find the Hicksian demands for good 1 and good 2 by solving the following expenditure minimization problem (EMP):

$$\min p_1x_1 + p_2x_2 \quad \text{subject to } kx_1^\alpha x_2^{1-\alpha} \geq u$$

$$x_1, x_2 \geq 0$$

FONC requires,

Lagrangian Function:

$$L = p_1x_1 + p_2x_2 + \lambda [u - kx_1^\alpha x_2^{1-\alpha}]$$

Taking first order necessary condition (FONC):

$$\frac{\partial L}{\partial x_1} = p_1 - \alpha (x_1^{\alpha-1}) \lambda k x_2^{1-\alpha} = 0$$

$$\Rightarrow p_1 = \lambda k \alpha x_1^{\alpha-1} x_2^{1-\alpha} \quad \text{--- (1)}$$

$$\frac{\partial L}{\partial x_2} = p_2 - \lambda k (1-\alpha) x_1^\alpha x_2^{-\alpha} = 0$$

$$\Rightarrow p_2 = \lambda k (1-\alpha) x_1^\alpha x_2^{-\alpha} \quad \text{--- (2)}$$

$$\frac{\partial L}{\partial \lambda} = u - kx_1^\alpha x_2^{1-\alpha} = 0$$

$$\Rightarrow kx_1^\alpha x_2^{1-\alpha} = u \quad \text{--- (3)}$$

where λ denotes the Lagrange multiplier of this minimization problem. In the case of interior solutions, the first order conditions above become

$$P_1 = \lambda k \alpha x_1^{\alpha-1} x_2^{1-\alpha} \quad \text{--- (4)}$$

$$P_2 = \lambda k (1-\alpha) x_1^\alpha x_2^{-\alpha} \quad \text{--- (5)}$$

$$k x_1^\alpha x_2^{1-\alpha} = u \quad \text{--- (6)}$$

Divide Eq. (4) by Eq. (5),

$$\frac{P_1}{P_2} = \frac{\lambda k \alpha x_1^{\alpha-1} x_2^{1-\alpha}}{\lambda k (1-\alpha) x_1^\alpha x_2^{-\alpha}}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{\alpha}{1-\alpha} \cdot x_1^{-1} x_2^{1-\alpha+\alpha}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{\alpha}{1-\alpha} \cdot \frac{x_2}{x_1}$$

$$\Rightarrow \frac{(1-\alpha) P_1 x_2}{\alpha P_2} = x_1 \quad \text{--- (7)}$$

Substituting Eq. (7) into Eq. (6),

$$k x_1^\alpha \left[\frac{(1-\alpha) P_1 x_2}{\alpha P_2} \right]^{1-\alpha} = u$$

$$\Rightarrow k x_1^\alpha x_2^{1-\alpha} \left(\frac{(1-\alpha) P_1}{\alpha P_2} \right)^{1-\alpha} = u$$

$$\Rightarrow x_1^{\alpha+1-\alpha} = \frac{u}{k \left[\frac{(1-\alpha) P_1}{\alpha P_2} \right]^{1-\alpha}} = u \times \frac{1}{k} \times \frac{(\alpha P_2)^{1-\alpha}}{((1-\alpha) P_1)^{1-\alpha}}$$

$$\Rightarrow x_1 = \frac{u \frac{1}{k} \frac{\alpha P_2}{(1-\alpha) P_1}}{\left[\frac{(1-\alpha) P_1}{\alpha P_2} \right]^{1-\alpha}}$$

Similarly, $x_2 = \left(\frac{(1-\alpha)P_1}{\alpha P_2} \right)^\alpha \cdot u \cdot \frac{1}{K}$

Thus, simultaneously solving for x_1 and x_2 , we find that the Hicksian demands for goods 1 and 2 are,

$$h_1(P, u) = \frac{1}{K} \left(\frac{\alpha P_2}{(1-\alpha)P_1} \right)^{1-\alpha} \cdot u \quad \text{and}$$

$$h_2(P, u) = \frac{1}{K} \left(\frac{(1-\alpha)P_1}{\alpha P_2} \right)^\alpha \cdot u$$

Therefore the expenditure function is

$$e(P, u) = P_1 \cdot h_1(P, u) + P_2 \cdot h_2(P, u)$$

$$\begin{aligned} &= P_1 \cdot \frac{1}{K} \left(\frac{\alpha P_2}{(1-\alpha)P_1} \right)^{1-\alpha} \cdot u + P_2 \cdot \frac{1}{K} \left(\frac{(1-\alpha)P_1}{\alpha P_2} \right)^\alpha \cdot u \\ &= \frac{1}{K} \left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} P_1^{1-(1-\alpha)} P_2^{1-\alpha} \cdot u + \frac{1}{K} \left(\frac{1-\alpha}{\alpha} \right)^\alpha P_1^\alpha P_2^{1-\alpha} \cdot u \\ &= \frac{1}{K} P_1^\alpha P_2^{1-\alpha} \cdot u \left[\left(\frac{\alpha}{1-\alpha} \right)^{1-\alpha} + \left(\frac{1-\alpha}{\alpha} \right)^\alpha \right] \end{aligned}$$

which is homogenous of degree one in prices, strictly increasing in u . In addition, $e(P, u)$ is strictly increasing in the prices of every good k , P_k and at a decreasing rate, thus satisfying concavity in prices.

Explaining the properties of Hicksian / compensated demand function

① Homogeneity of degree zero in prices: $h(p, u) = h(\alpha p, u)$, for any p, u and $\alpha > 0$.

It is straight-forward to show that the Hicksian demands satisfy homogeneity of degree zero in prices, since $\nearrow$ all ^{increasing} prices by the same factor $\alpha > 0$ keeps both demand functions unaffected

$$\begin{aligned} h_1(p, u) &= \frac{1}{K} \left(\frac{\alpha A p_2}{(1-\alpha) A p_1} \right)^{1-\alpha} \cdot u \\ &= \frac{1}{K} \left(\frac{\alpha p_2}{(1-\alpha) p_1} \right)^{1-\alpha} \cdot u \end{aligned}$$

$$\begin{aligned} h_2(p, u) &= \frac{1}{K} \left(\frac{(1-\alpha) A p_1}{\alpha A p_2} \right)^{\alpha} \cdot u \\ &= \frac{1}{K} \left(\frac{(1-\alpha) p_1}{\alpha p_2} \right)^{\alpha} \cdot u \end{aligned}$$

If $\tilde{x} \in h(p, u)$ is a solution w/p_2 to the problem, $\min_{x \geq 0} p \cdot x$. 
then it is also solution to the problem, $\min_{x \geq 0} \alpha p \cdot x$.
A common change in all prices does not alter the slope of the consumer's budget line. $\tilde{x}$ is a solution to the EMP the price vector $p = (p_1, p_2)$. Increase prices by factor α , $p' = \alpha p$, $p'_1 = \alpha p_1$, $p'_2 = \alpha p_2$. Downward (parallel) shift in the budget line i.e. the slope of the budget line is unchanged. But I have to reach u level u to satisfy EMP constraint. $\therefore$ more to buy bundle $\tilde{x}^*(x_1^*, x_2^*)$ i.e. $p'_1 x_1^* + p'_2 x_2^* > p_1 x_1^* + p_2 x_2^*$. Hence, $h(p, u) =$

② Convexity: If the preference relation is convex, then $h(p, u)$ is a convex set. (This property is also related to the following uniqueness property).

③ Uniqueness: If the preference relation is strictly convex, then $h(p, u)$ contains a single element.

The Hicksian demands are also unique, since for a given price vector, a given utility level, u and a specific value for parameters, α and β , we can obtain a precise

quantity demanded for good 1 and good 2.

④ No excess utility:

For any optimal consumption bundle $x \in h(p, u)$ that solves the EMP, the utility level satisfies $u(x) = u$.

Finally, the "no excess utility" property holds, since, in evaluating the Cobb-Douglas utility function at the Hicksian demands, we obtain the exact utility level u that we sought to reach in the EMP. In particular,

$$\begin{aligned} u(h_1(p, u), h_2(p, u)) &= h_1(p, u)^\alpha \cdot h_2(p, u)^{1-\alpha} \\ &= \left[\frac{1}{k} \left(\frac{\alpha p_2}{(1-\alpha)p_1} \right)^{1-\alpha} \cdot u \right]^\alpha \cdot \left[\frac{1}{k} \left(\frac{(1-\alpha)p_1}{\alpha p_2} \right)^\alpha \cdot u \right]^{1-\alpha} \\ &= \left(\frac{1}{k} \right)^\alpha \left(\frac{\alpha p_2}{(1-\alpha)p_1} \right)^{(1-\alpha)\alpha} u^\alpha \cdot \left(\frac{1}{k} \right)^{1-\alpha} \left(\frac{(1-\alpha)p_1}{\alpha p_2} \right)^{(1-\alpha)(1-\alpha)} u^{1-\alpha} \\ &= \left(\frac{1}{k} \right)^{\alpha + 1 - \alpha} \frac{\alpha^{1-\alpha} (1-\alpha)^\alpha}{\alpha^{1-\alpha} (1-\alpha)^\alpha} \frac{(1-\alpha)^\alpha}{\alpha^{1-\alpha}} \cdot u^{\alpha + 1 - \alpha} \\ &= \frac{1}{k} (\alpha p_2)^\alpha \cdot (1-\alpha)^\alpha p_1^{1-\alpha} \cdot u \end{aligned}$$

$$\begin{aligned} &\alpha - \alpha^2 - \alpha + \alpha^2 = -\alpha + \alpha^2 + \alpha - \alpha^2 \\ &= \frac{1}{k} (\alpha p_2)^\alpha \cdot (1-\alpha)^\alpha p_1^{1-\alpha} \cdot u \\ &= \frac{1}{k} (\alpha p_2)^\alpha \left[(1-\alpha)^\alpha p_1^{1-\alpha} \right] \cdot u \\ &= \frac{1}{k} \cdot u \quad \text{as required.} \end{aligned}$$

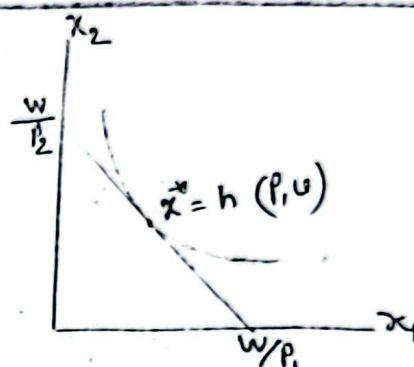
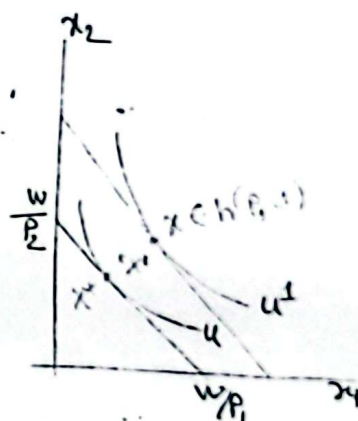


Fig: Convexity and uniqueness



Suppose bundle $x \in h(p, u)$ for which consumer obtains a utility level $u(x) = u$, which is higher than the utility level u he must reach when solving EMP.

But, another bundle $x' = \alpha x$, where $\alpha \in (0, 1)$, very close to x ($\alpha \rightarrow 1$), for which $u(x') > u$.

Bundle x' is cheaper than x since it contains fewer units of all goods and exceeds the minimal utility level u that the consumer must reach in his EMP.

We can repeat the argument until reaching bundle x^* .